



Large Intestine & Anal Canal

Histology GIT Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty, 11 case-based. Attempt every question before checking the answer key on the last page.

● 4 Easy

● 10 Medium

● 6 Hard

1.

EASY

Which structure is characteristic of the large intestine but not the small intestine?

- A. Brunner's glands
- B. Villi
- C. Peyer's patches
- D. Taeniae coli
- E. Plicae circulares

2.

EASY

The outer longitudinal muscle of the colon is gathered into how many taeniae coli?

- A. Six
- B. Four
- C. One
- D. Two
- E. Three

3.

EASY

Which crypt cells secrete mucin for lubrication in the large intestine?

- A. Stem cells
- B. Paneth cells
- C. Enteroendocrine cells
- D. Absorptive cells
- E. Goblet cells

4.

EASY

Approximately how long is the anal canal?

- A. 1.5 m
- B. 25 cm
- C. 4 cm
- D. 10 cm
- E. 1 cm

5.

MEDIUM

CASE

A mucosal section shows no villi and no folds. Long, straight crypts are packed with goblet cells and the lamina propria contains many lymphatic nodules.

Which organ is this?

- A. Pyloric stomach
- B. Ileum
- C. Duodenum
- D. Large intestine
- E. Jejunum

6.

MEDIUM

The outer longitudinal layer of the colon is:

- A. replaced by an oblique layer
- B. a uniformly thick complete sheet
- C. absent from the colon wall
- D. gathered into three thick bands with thin fibres between
- E. gathered into two thick bands

7.

MEDIUM

CASE

A section shows the histology of the large intestine, but the outer longitudinal layer forms a complete sheet without bands, and the outermost layer is an adventitia.

Which segment is this?

- A. Cecum
- B. Rectum
- C. Sigmoid colon
- D. Transverse colon
- E. Ileum

8.

MEDIUM

Which pairing of segment and outer covering is correct?

- A. Transverse colon — adventitia; sigmoid colon — adventitia; rectum — serosa
- B. Cecum — adventitia and serosa; ascending colon — serosa; rectum — serosa
- C. Cecum — serosa; ascending colon — adventitia and serosa; rectum — adventitia
- D. Cecum — adventitia; ascending colon — serosa; rectum — serosa
- E. Ascending colon — serosa only; descending colon — serosa only; rectum — adventitia

9.

MEDIUM

Compared with the small intestine, the lamina propria of the large intestine contains:

- A. Peyer's patches only
- B. Brunner's glands
- C. fewer lymphatic nodules
- D. more lymphatic nodules
- E. villus cores

10.

MEDIUM

CASE

A biopsy is taken from the upper 15 mm of the anal canal, where 6–12 longitudinal folds project into the lumen.

Which epithelium lines this zone?

- A. Keratinized stratified squamous
- B. Columnar epithelium
- C. Non-keratinized stratified squamous
- D. Pseudostratified ciliated
- E. Transitional

11.

MEDIUM

CASE

A section from the lowest 8–10 mm of the anal canal shows hair follicles, sebaceous glands and sweat glands.

What is this tissue?

- A. True skin
- B. Rectal mucosa with crypts
- C. Non-keratinized stratified squamous mucosa
- D. Transitional epithelium
- E. Simple columnar mucosa

12.

MEDIUM

The internal anal sphincter is:

- A. thickened circular smooth muscle under autonomic control
- B. the muscularis mucosae of the canal
- C. thickened longitudinal skeletal muscle
- D. striated muscle under voluntary control
- E. the outer layer forming taeniae coli

13.

MEDIUM

CASE

In the submucosa of the anal canal, a prominent venous plexus lies just above the pectinate line.

Which plexus is this?

- A. External hemorrhoidal plexus
- B. Internal hemorrhoidal plexus
- C. Lymphatic plexus
- D. Meissner's plexus
- E. Auerbach's plexus

14.

MEDIUM

The appendix differs from the colon in having:

- A. no lymphatic nodules
- B. more, longer glands and three taeniae coli
- C. villi on its surface
- D. a wide regular lumen
- E. fewer, shorter glands and no taeniae coli

15.

HARD

CASE

Three biopsies from one anal canal. A: columnar epithelium with longitudinal folds. B: non-keratinized stratified squamous epithelium. C: skin with hair follicles and sweat glands.

Which order runs from proximal to distal?

- A. A, B, C
- B. A, C, B
- C. C, B, A
- D. B, C, A
- E. B, A, C

16.

HARD

CASE

Section A shows villi with a brush border. Section B has no villi; long straight crypts and numerous goblet cells reach the muscularis mucosae.

Which section is small intestine, which large intestine, and why?

- A. A is small intestine; B is large intestine, because B has Paneth-cell granules and Peyer's patches
- B. Both are small intestine; the crypts differ only in length
- C. A is large intestine; B is small intestine, because its crypts are long and straight
- D. Both are large intestine; villi are present throughout
- E. A is small intestine; B is large intestine, whose crypts are long and straight and which has no villi

17.

HARD

CASE

A venous plexus lies in the submucosa just above the dentate line, beneath columnar epithelium. Another lies near the lower end of the canal, beneath the skin-lined zone.

Which statement is correct?

- A. Both are internal hemorrhoidal plexuses
- B. Upper = internal plexus (columnar zone); lower = external plexus near the canal end
- C. Both are external plexuses below the anal verge
- D. Upper = external plexus; lower = internal plexus
- E. Upper = Meissner's plexus; lower = Auerbach's plexus

18.

HARD

CASE

A patient cannot voluntarily hold back defecation, although resting sphincter tone is preserved.

Which structure is damaged, and what is its composition?

- A. External anal sphincter; striated muscle under voluntary control
- B. External anal sphincter; smooth muscle under autonomic control
- C. Internal anal sphincter; striated muscle under voluntary control
- D. Taeniae coli; smooth muscle
- E. Internal anal sphincter; smooth muscle under autonomic control

19.

HARD

CASE

A blind-ending tube, an evagination of the cecum about 5–10 cm long, has a small narrow irregular lumen and abundant lymphatic nodules with germinal centres in its wall.

Which additional finding would confirm the diagnosis?

- A. Three thick taeniae coli and long straight crypts
- B. Villi over the mucosa and submucosal Brunner's glands
- C. Fewer, shorter glands; no taeniae coli; outer longitudinal + inner circular muscle
- D. A wide regular lumen with regular mucosal folds
- E. Hair follicles and sweat glands in the lining

20.

HARD

CASE

From the lumen: a mucosa without folds, a submucosa, an inner circular layer, an outer longitudinal layer gathered into three thick bands, and a serosa.

Which sequence of wall layers and covering fits this colon segment?

- A. No folds, complete longitudinal sheet, adventitia: the transverse colon
- B. Villi, two smooth layers, adventitia: the rectum
- C. No mucosal folds, three thick bands, serosa: cecum, transverse or sigmoid colon
- D. Plicae circulares, inner circular, outer longitudinal, serosa: the jejunum
- E. Skin-lined mucosa, external sphincter: the sigmoid colon

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	D	The outer longitudinal layer gathered into three thick bands (taeniae coli) marks the colon.
2	E	Three thick longitudinal bands, with thin longitudinal fibres between them.
3	E	Goblet cells contain mucin granules and act as a lubricant.
4	C	The anal canal is about 4 cm long, connecting the rectum to the anal opening.
5	D	Deep, long, straight crypts with numerous goblet cells and no villi identify the large intestine.
6	D	In the colon the outer longitudinal layer forms three taeniae coli.
7	B	Large-intestine histology without taeniae coli and with an adventitia = rectum.
8	C	Intraperitoneal cecum, transverse and sigmoid colon: serosa; retroperitoneal ascending / descending: adventitia & serosa; rectum: adventitia.
9	D	The lecture states the large intestine lamina propria has more lymphatic nodules than the small intestine.
10	B	The mucous membrane of the upper 15 mm of the anal canal is columnar (with the anal columns).
11	A	The lowest 8–10 mm of the anal canal is lined by true skin with hair follicles, sebaceous and sweat glands.
12	A	The circular muscle thickens to form the internal sphincter (smooth, autonomic); the external sphincter is striated and voluntary.
13	B	The internal hemorrhoidal plexus is located above the pectinate line.
14	E	Appendix: fewer shorter glands, no taeniae coli, narrow irregular lumen, abundant nodules.
15	A	Upper 15 mm columnar → lower 15 mm non-keratinized stratified squamous → lowest 8–10 mm true skin.
16	E	Villi with a brush border = small intestine; no villi with long straight crypts and numerous goblet cells = large intestine.
17	B	Internal hemorrhoidal plexus above the pectinate line; external plexus near the lower end of the canal.
18	A	Voluntary control = external sphincter = striated muscle; the intact internal sphincter is smooth and autonomic.
19	C	The appendix has fewer shorter glands, no taeniae coli, and a muscularis externa arranged as outer longitudinal and inner circular.
20	C	Three taeniae coli plus serosa = intraperitoneal colon (cecum, transverse, sigmoid); the rectum has no taeniae and an adventitia.

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology SGL 3, Dr Hero K. Mustafa
Written from the lecture slides only – check any answer you doubt against your lecture notes.